

# Ivan Plyushchenko

## Postdoctoral Fellow

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University of Haifa, Department of Marine Biology



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### Education

|         |                                              |                                   |
|---------|----------------------------------------------|-----------------------------------|
| 2017-22 | PhD (Chemistry)<br>Moscow, Russia            | Lomonosov Moscow State University |
| 2011-17 | MSc (Chemistry), GPA: 93.4<br>Moscow, Russia | Lomonosov Moscow State University |

### Key Publications

|      |                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | Hopomics: Humulus lupulus Brewing Cultivars Classification Based on LC-MS Profiling and Nested Feature Selection<br>Metabolites                                       |
| 2022 | Omics Untargeted Key Script: R-Based Software Toolbox for Untargeted Metabolomics with Bladder Cancer Biomarkers Discovery Case Study<br>Journal of Proteome Research |
| 2021 | Deep learning for retention time prediction in reversed-phase liquid chromatography<br>Journal of Chromatography A                                                    |
| 2020 | An approach for feature selection with data modelling in LC-MS metabolomics<br>Analytical Methods                                                                     |

### Grants and Awards

|         |                                                                                                                                                                                                         |             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 2018-22 | Russian Foundation for Basic Research<br>3 grants                                                                                                                                                       | Participant |
| 2017-21 | Russian Science Foundation<br>2 grants                                                                                                                                                                  | Participant |
| 2026-27 | Institutional Postdoctoral Fellowship<br>University of Haifa, Israel                                                                                                                                    |             |
| 2025-26 | Institutional Postdoctoral Fellowship<br>University of Haifa, Israel                                                                                                                                    |             |
| 2022    | Winner of the competition of works contributing to the solution of problems of the Development Program of Moscow State University in the nomination 'Outstanding scientific articles'<br>Moscow, Russia |             |
| 2022    | Competition of the Russian Academy of Sciences of scientific works of young scientists in chromatography in honor of the 150th anniversary of the birth M.S. Tsvet<br>Moscow, Russia                    |             |
| 2019    | The Association for Mass Spectrometry & Advances in the Clinical Lab (MSACL) Conference EU Travel Grant<br>Salzburg, Austria                                                                            |             |
| 2018    | International Mass Spectrometry Conference Fellowship<br>Florence, Italy                                                                                                                                |             |
| 2017    | The Association for Mass Spectrometry & Advances in the Clinical Lab (MSACL) Conference EU Travel Grant<br>Salzburg, Austria                                                                            |             |

## Professional Experience

- 2025.01-present Postdoctoral Fellow  
University of Haifa, Department of Marine Biology
- 2022.12-2024.12 Graduate Researcher  
Technion - Israel Institute of Technology, Biology Faculty
- 2022.03-2022.11 Postdoctoral Research Fellow  
Lomonosov Moscow State University, Chemistry Department
- 2017.07-2022.02 Junior Research Associate  
Lomonosov Moscow State University, Chemistry Department
- 2016.05-2017.06 Lab Technician  
Lomonosov Moscow State University, Chemistry Department
- 2015.05-2018.06 Analytical Chemist  
Analytical centre of Lomonosov Moscow State University

## Skills & Proficiencies

### Instruments & techniques:

- MS: SCAN, SIM, MRM, DDA, DIA, PASEF, TIMS
- LC: RP, HILIC
- Flash Chromatography: CombiFlash NextGen 300 (Teledyne)
- Nano LC: Evosep One (Evosep)
- LC-MS: IT-TOF (Shimadzu); Orbitrap Exploris 120, Orbitrap Q Exactive (Thermo); QTRAP 3200 (Sciex); 6470 (Agilent); timsTOF Pro 2 (Bruker)
- LC-DAD: 1100, 1200, 1290 Infinity (Agilent); Ultimate 3000 (Dionex); LC 20 (Shimadzu); Vanquish (Thermo)
- LC-FLD: 1200 (Agilent); Ultimate 3000 (Dionex)
- CE-UV: G1600AX, 7100 (Agilent)
- GC-MS: GCMS-QP 2010 with SHS (Shimadzu); 7890 B (Agilent); Trace 1610 & ISQ 7610 with SHS (Thermo)
- GC-FID: GC-2010 (Shimadzu)
- MALDI: autoflex maX for Imaging & Dried Drop (Bruker)

### Basic lab skills:

- Extraction from liquid & solid media
- SPE, LLE, Soxhlet extraction, centrifugation, membrane filtering
- UV/Vis analysis, pH measurement
- Preparation of calibration curves, QC/QA programs performing, daily maintenance
- Maintenance of mammalian cell lines and bacterial cultures

### Programming, Software & Bioinformatics tools:

- R language independent user
- tidyverse, tidymass, RforMassSpectrometry families of packages
- xcms, CAMERA, IPO, MSnbase, MSstats, Biostrings, seqinr, rcdk, caret, H2O, keras, xgboost, ggplot2, dplyr, data.table, RMarkdown, knitr, plotly, Shiny, usethis, devtools, roxygen2, testthat, pkgdown
- R packages / Shiny tools list: MetCorR URL, MetaboCensoR URL, Metabocano URL, qry4ms URL, OmniPeak URL
- developed tool list: URL
- full list of R packages used for metabolomics processing: URL
- full list of tools used for MS processing: URL
- GUI software: MetaboAnalyst, MS-DIAL, MZmine, MS-FINDER, Maven, SIR-IUS, CFM-ID, MetFrag, GNPS, Open Babel, PathVisio, Cytoscape, FragPipe, DIA-NN
- Vendor Software: LabSolutions (Shimadzu); XCalibur, FreeStyle, Compound Discoverer (Thermo); OpenLab, MassHunter, ChemStation (Agilent); Compass HyStar, OtofControl (Bruker)

## Certificates & Courses

|      |                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Introduction to Metabolomics 02780958<br>Technion, Faculty of Medicine <ul style="list-style-type: none"><li>• 24 hrs</li></ul>                                                                                                                            |
| 2019 | Data Science 201: Going Further With R: Tackling Clinical Laboratory Data Manipulation and Modeling<br>The Association for Mass Spectrometry & Advances in the Clinical Lab (MSACL) Conference EU <ul style="list-style-type: none"><li>• 16 hrs</li></ul> |
| 2019 | Basics of Programming in R<br>Bioinformatics Institute, Russia <ul style="list-style-type: none"><li>• 6 hrs</li></ul>                                                                                                                                     |
| 2019 | Machine Learning<br>Omsk State Technical University, Russia <ul style="list-style-type: none"><li>• 7 hrs</li></ul>                                                                                                                                        |
| 2019 | Data Analysis in R. Part 2<br>Bioinformatics Institute, Russia <ul style="list-style-type: none"><li>• 7 hrs</li></ul>                                                                                                                                     |
| 2018 | Data Analysis in R<br>Bioinformatics Institute, Russia <ul style="list-style-type: none"><li>• 6 hrs</li></ul>                                                                                                                                             |
| 2017 | Metabolomics 202: Approaches, Applications and Challenges<br>The Association for Mass Spectrometry & Advances in the Clinical Lab (MSACL) Conference EU <ul style="list-style-type: none"><li>• 16 hrs</li></ul>                                           |

## Teaching and Mentorship

|         |                                                                                                                                          |
|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| 2018-22 | Mentoring undergraduates students of Lomonosov Moscow State University, Faculty of Chemistry, including two students' thesis supervision |
| 2018-20 | Practical and theoretical classes on analytical chemistry for students of Lomonosov Moscow State University, Faculty of Biology          |
| 2016-19 | Agilent advanced training courses at the Analytical centre of Lomonosov Moscow State University in LC, LC-MS                             |

## Publication List

- 2026 Deciphering temporal antifungal dynamics of a rare actinomycete via integrated omics  
**mSystems**
- 2023 Glyphosate Effects on Some Characteristics of Biological Activity and Phytotoxicity of Soddy-Podzolic Soil in a Short-Term Model Experiment  
**Eurasian Soil Science**
- 2022 Hopomics: Humulus lupulus Brewing Cultivars Classification Based on LC-MS Profiling and Nested Feature Selection  
**Metabolites**
- 2022 Omics Untargeted Key Script: R-Based Software Toolbox for Untargeted Metabolomics with Bladder Cancer Biomarkers Discovery Case Study  
**Journal of Proteome Research**
- 2021 Algorithm of Combining Chromatography–Mass Spectrometry Untargeted Profiling and Multivariate Analysis for Identification of Marker Substances in Samples of Complex Composition  
**Inorganic Materials**
- 2021 Deep learning for retention time prediction in reversed-phase liquid chromatography  
**Journal of Chromatography A**
- 2021 Development of ELISA formats for polymyxin B monitoring in serum of critically ill patients  
**Journal of Pharmaceutical and Biomedical Analysis**
- 2021 Fused 1,2-Diboraoxazoles Based on closo-Decaborate Anion–Novel Members of Diboroheterocycle Class  
**Molecules**
- 2021 Untargeted and targeted analysis of sarin poisoning biomarkers in rat urine by liquid chromatography and tandem mass spectrometry  
**Analytical and Bioanalytical Chemistry**
- 2020 An approach for feature selection with data modelling in LC-MS metabolomics  
**Analytical Methods**
- 2020 Monoammonium Phosphate Effects on Glyphosate in Soils: Mobilization, Phytotoxicity, and Alteration of the Microbial Community  
**Eurasian Soil Science**
- 2020 The Degradation of Glyphosate and Its Effect on the Microbial Community of Agro-Sod–Podzolic Soil under Short-Term Model Experiment Conditions  
**Moscow University Soil Science Bulletin**
- 2019 Use of Chemometric Methods of Data Analysis for the Identification and Typification of Petroleum and Petroleum Products  
**Journal of Analytical Chemistry**
- 2018 Identification of Spillages of Semi-Volatile Hydrocarbon Fuels in Soils by Gas Chromatography–Mass Spectrometry  
**Journal of Analytical Chemistry**
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## Conferences

- 2026 Unraveling the Complexity of Microalgal EPS: A Multi-Analytical Approach to Polysaccharide Composition and Characterization ASMS 74th, San Diego, USA
- 2026 Heat, Halogens, and Holobionts: LC-MS/MS Insights into Seasonal Metabolite Dynamics in *Asparagopsis taxiformis* ASMS 74th, San Diego, USA
- 2026 Mass spectrometry reveals metabolic rewiring in *Bacillus subtilis* following loss of surfactin and plipastatin ASMS 74th, San Diego, USA
- 2023 Untargeted urinary metabolomics for identification of bladder cancer biomarkers using HPLC-MS Euroanalysis XXI, Geneva, Switzerland
- 2022 Application of gradient boosting machine for signal processing in LC-MS metabolomics 13th Winter Symposium on Chemometrics, Russia
- 2022 Untargeted metabolomics study of *Humulus lupulus* brewing cultivars, for genetic origin classification task 13th Winter Symposium on Chemometrics, Russia
- 2021 Describing metabolome diversity between *Humulus lupulus* genetic origin groups using UHPLC-MS/MS The 69th Annual Conference on Mass Spectrometry of MSSJ, Japan
- 2020 *Humulus lupulus* LC-MS untargeted profiling study for geographic origin classification task 4th International Symposium on Phytochemicals in Medicine and Food, Xi'an, China
- 2019 Comparison of the kinetics of dyes degradation of handwritten strokes subjected to different types of artificial aging and studied using chromatography mass-spectrometry and statistical data processing 48th International Symposium on High-Performance Liquid Phase Separations and Related Techniques, Milan, Italy
- 2019 Bioassay Classification Study via LC-MS and Machine Learning in Conjunction with Dimensionality Reduction MSACL 2019 EU, Salzburg, Austria
- 2018 Typical LC-MS metabolomics workflow for profiling urine samples of patients with colorectal cancer XXII International Mass Spectrometry Conference, Florence, Italy
- 2017 Simple & Robust Approach in Urinary Metabolomics Based on UPLC-MS for Preoperative Colorectal Cancer Diagnostics MSACL 2017 EU, Salzburg, Austria
- 2016 Complexation of sulfo-beta-cyclodextrin with fenoterol. Electrophoretic and spectroscopic study VIIIth International Symposium Design and Synthesis of Supramolecular Architectures, Kazan, Russia

## Current Journal Roles

|                                                     |               |
|-----------------------------------------------------|---------------|
| Biomedical Chromatography                           | Peer Reviewer |
| Journal of Analytical Chemistry                     | Peer Reviewer |
| BMC Medicine                                        | Peer Reviewer |
| Journal of Food Composition and Analysis            | Peer Reviewer |
| Scientific Reports                                  | Peer Reviewer |
| PLOS Computational Biology                          | Peer Reviewer |
| BMC Infectious Diseases                             | Peer Reviewer |
| PLOS One                                            | Peer Reviewer |
| Proteome Science                                    | Peer Reviewer |
| Journal of Proteome Research                        | Peer Reviewer |
| Scientific Data                                     | Peer Reviewer |
| BMC Bioinformatics                                  | Peer Reviewer |
| BMC gastroenterology                                | Peer Reviewer |
| Nutrition & metabolism                              | Peer Reviewer |
| npj Systems Biology and Applications                | Peer Reviewer |
| Journal of Computational Chemistry                  | Peer Reviewer |
| BMC Cancer                                          | Peer Reviewer |
| ACS Omega                                           | Peer Reviewer |
| Metabolomics                                        | Peer Reviewer |
| Journal of Biologically Active Products from Nature | Peer Reviewer |
| Communications Medicine                             | Peer Reviewer |
| Discover Artificial Intelligence                    | Peer Reviewer |
| BMC Research Notes                                  | Peer Reviewer |
| BMC Oral Health                                     | Peer Reviewer |